Android Mobile Application Development (Level-4)

Complete Assessment Bank with Fully Integrated Answers & Solutions

Zero Duplicate Questions • Clean Format • 3 Practical Jobs, 37 Solved MCQs, 14 Solved Short Answers & 10 Solved Oral Questions

Part 1: Practical Assessment & Job Specification Sheets

Units of Competency (UoCs) Covered:

- `GU-02-L2-V1`: Apply occupational health and safety (OHS) practices
- `QU-ICT-AMAD-01-L4-V1`: Work with Kotlin Basics
- `QU-ICT-AMAD-02-L4-V1`: Work with OOP and Design Patterns
- `QU-ICT-AMAD-03-L4-V1`: Work with Android Basics
- `QU-ICT-AMAD-04-L4-V1`: Manage Database (SQLite)
- `QU-ICT-AMAD-05-L4-V1`: API Integration & Background Services

Standard Working Procedure / Steps:

1. Follow OSH guidelines and equip PPE.
2. Check power, internet, and start Android Studio.
3. Name root folder: `RegistrationNo_FullName`.
4. Read specification sheets and requirements.
5. Implement architecture (MVVM), UI, logic, & permissions.
6. Test on emulator/real device and clean workplace.
7. Zip project directory and submit to assessor.

Job 01 – REST API Consumer Application

TIME: 1 HOUR **PROJECT: NAME_ID_REST01**

API Endpoints: `https://api.escuelajs.co/api/v1/products` | `https://fakeapi.platzi.com/en/rest/products`

1. Create Android Studio project named **Name_ID_REST01** using Kotlin.
2. Build `ProductActivity` featuring a `RecyclerView` and custom item layout XML.
3. Add dependencies for REST client (Retrofit + Gson Converter) & Image loading library (Glide / Coil) in `app/build.gradle`.
4. Declare `<uses-permission android:name="android.permission.INTERNET"/>` in `AndroidManifest.xml`.
5. Create data model (e.g. `ProductItem`: id, title, price, description, images) and Retrofit API interface.
6. Make asynchronous network calls, parse JSON response, and render product cards with image caching.
7. Handle offline state gracefully by displaying an error state with a working **Retry** button.

Job 02 – User Profile Registration (SQLite)

TIME: 1 HOUR **PROJECT: USERPROFILEREGISTRATION**

1. Create project named **UserProfileRegistration** using MVVM architecture (ViewModel + LiveData).
2. Build launcher `WelcomeActivity` directing users to view or add profiles.
3. Create `ProfileListActivity` with a `RecyclerView` to display all registered profiles and total count.
4. Build `AddProfileActivity` to input: Name, Email, Date of Birth, District, and Mobile Number.
5. Implement local database via SQLite / Room to support full CRUD operations (Create, Read, Update, Delete).
6. Provide **Edit Profile** (update record) and **Delete Profile** actions with immediate UI update.
7. Create `SingleProfileActivity` to inspect full individual user details.

Job 03 – Real-Time Location Sharing (Firebase & Maps)

TIME: 2 HOURS **PROJECT: LOCATIONSHARINGAPP**

1. Create **LocationSharingApp** project and link to Firebase (Auth + Cloud Firestore).
2. Configure Firestore security rules to allow authenticated read/write operations only.
3. Add dependencies for Firebase Auth, Cloud Firestore, Google Play Services Location, and Maps SDK.
4. Add Location permissions (`ACCESS_FINE_LOCATION`) & Google Maps API key in `AndroidManifest.xml`.
5. Build `AuthActivity` for registration/login, then fetch GPS coordinates via `FusedLocationProviderClient`.
6. Save `AppUser` data object (`userId`, `userEmail`, `displayName`, `latitude`, `longitude`) to Firestore.
7. Create `FriendListActivity` with a Floating Options Menu (*My Profile*, *Show on Map*, *Logout*).
8. Build `GoogleMapActivity` placing map markers for all registered users with interactive info window snippets.

Required Tools, Materials, and Personal Protective Equipment (PPE)

S/N	Category / Name	Specification	Unit	Qty
01	Ergonomic Chair	Wood and foam adjustable ergonomic office chair	No	1
02	Eye Protective Glass	Metal & Glass anti-glare protective spectacles	No	1
03	Rubber Shoes / Mat	Rubber antistatic safety gear	Pair	1
04	PC / Laptop Workstation	Min Core i3/i5 7th Gen+, 16GB RAM, SSD Storage	Set	1

S/N	Category / Name	Specification	Unit	Qty
05	Development Suite	Android Studio (Latest stable) + JDK 11/17	Set	1
06	Test Device / Emulator	Android Emulator / Physical Device (API 21+)	Set	1
07	Network & Accounts	High-speed Internet connection + Google/Firebase Account	Acc	1

Part 2: Multiple Choice Questions (Zero Duplicates - Fully Solved)

1. Identify the primary programming language Android system development was originally based upon: ANSWER: A

A) Java B) C++
C) Python D) C

2. What does the acronym AAPT stand for in the Android build toolchain? ANSWER: C

A) Android Activity Packaging Tool B) Android Action Packaging Tool
C) Android Asset Packaging Tool D) Android Application Process Tool

3. Which of the following is NOT a native core feature of Android? ANSWER: C

A) Connectivity B) Storage
C) Slide Mobango D) Multi-touch

4. What is the primary purpose of a Content Provider in Android? ANSWER: B

A) For storing the data in the database exclusively **B) For sharing the data between applications**
C) For sending the data from an application to another application D) None of the above

5. Which of the following is the parent class of Activity? ANSWER: C

A) Context B) Object
C) ContextThemeWrapper D) None of the above

6. In which of the following tabs is an error shown in Android Studio? ANSWER: D

A) CPU B) Memory
C) ADB Logs **D) Logcat**

7. In Kotlin, which keyword/symbol is used to define a lambda expression? ANSWER: B

A) lambda **B) ->**
C) => D) _>

8. Which of the following is a conditional statement/expression in Kotlin? ANSWER: C / D

A) for B) while
C) if **D) when**

9. What is the Android Application Package / App Bundle format primarily used for? ANSWER: A

A) Distributing finalized app versions B) Storing source code
C) Running app tests D) Generating APK files

10. Which lifecycle method is called when an Android Service is destroyed? ANSWER: C

A) onCrash() B) onStop()
C) onDestroy() D) onServiceEnd()

11. What is the purpose of the GoogleMap.OnMarkerClickListener interface in Android's Map API? ANSWER: B

A) It plays audio **B) It handles user clicks on map markers**
C) It manages device contacts D) It displays images

12. Which HTTP status code indicates that a request was successful and the server has fulfilled it? ANSWER: A

A) 200 OK B) 404 Not Found
C) 500 Internal Server Error D) 401 Unauthorized

13. Which library is commonly used in Android development to make HTTP requests when working with external APIs? **ANSWER: A**

A) Retrofit
B) Gson
C) Picasso
D) Dagger

14. What class is used to display a map fragment in an Android app's user interface? **ANSWER: C / A**

A) MapFragment
B) MapDisplayFragment
C) MapView / SupportMapFragment
D) MapActivity

15. To use FCM in your mobile app, you need to integrate which SDK? **ANSWER: A**

A) Firebase SDK
B) Android Studio SDK
C) iOS SDK
D) Microsoft Visual Studio SDK

16. Which of the following is the topmost layer of Android architecture? **ANSWER: C**

A) System Libraries and Android Runtime
B) Linux Kernel
C) Applications
D) Applications Framework

17. SQLite database engine is written in: **ANSWER: D / A**

A) C
B) C#
C) C++
D) ANSI-C

18. Which of the following is NOT a constraint in SQL? **ANSWER: D**

A) Primary key
B) Not Null
C) Check
D) Union

19. In order to sort the records according to the columns, which clause is used? **ANSWER: C**

A) HAVING
B) GROUP BY
C) ORDER BY
D) None of these

20. What is the purpose of the "else" block in an "if-else" statement in Kotlin? **ANSWER: B**

A) It is used to define another condition
B) It executes if the "if" condition is false
C) It is used to terminate the program
D) It is optional and can be wrong

21. Which of the following represents a Many-to-Many association? **ANSWER: A**

A) A student enrolling in multiple courses, and a course having multiple students
B) A user placing multiple orders
C) An employee having one manager
D) A library book having multiple copies

22. In Kotlin, which access modifier is used to restrict the visibility of class members strictly within the class? **ANSWER: B**

A) protected and internal
B) private
C) public and internal
D) private and internal

23. In Kotlin, how do you define a getter and setter for a class property? **ANSWER: A**

A) By using the "get" and "set" keywords beneath property
B) By using a separate function for each
C) By using "get" and "set" methods within the class
D) None of above

24. Which keyword is used to declare a function in Kotlin? **ANSWER: A**

A) fun
B) val
C) function
D) FunOrder

25. What is the Android Debug Bridge (ADB) used for in the context of mobile app development? **ANSWER: B**

A) To design user interfaces
B) To install, debug, and communicate with Android devices/ emulators
C) To monitor app battery usage exclusively
D) To create icon assets

26. What happens if a registered BroadcastReceiver is not unregistered in an Android app? ANSWER: A

A) It causes Activity memory leaks and redundant background processing
B) The receiver will keep processing broadcasts indefinitely without issue
C) The app will receive double the number of broadcasts
D) The receiver will only work when the app is in background

27. Which tool is used to generate a digital certificate/keystore for Android app signing? ANSWER: B

A) Android Studio UI
B) keytool / Keystore Utility
C) KeyPair
D) Gradle

28. What is the primary purpose of a Broadcast Receiver in Android? ANSWER: B

A) To send data between activities
B) To listen for and respond to system-wide broadcast messages
C) To handle user touch events
D) To manage database operations

29. To declare a Broadcast Receiver in your Android app, which component do you define it in? ANSWER: B

A) Activity
B) AndroidManifest.xml
C) Application class
D) Gradle script

30. In Android, mini activities or modular sub-screens are also known as: ANSWER: C

A) Adapter
B) Activity
C) Fragment
D) None Above

31. Identify the dialogue class in Android among the following: ANSWER: D

A) DatePickerDialog
B) AlertDialog
C) ProgressDialog
D) All of the above

32. Which of the following is the first callback method that is invoked by the system during an activity life-cycle? ANSWER: B

A) onClick() method
B) onCreate() method
C) onStart() method
D) onRestart() method

33. Which of the following data type is most appropriate for storing 252 characters? ANSWER: B / A

A) Text
B) TINYTEXT / VARCHAR(255)
C) BLOB
D) Binary

34. In SQLite files are _____ directly to disk: ANSWER: C

A) Read
B) Written
C) Both A and B (Read and Written)
D) None of the above

35. SQLite supports many of the features of SQL, and is fast. However, _____ are not supported: ANSWER: A

A) Stored Procedures
B) Indexes
C) Failed transactions
D) Transactions

36. A table can be _____ with the ALTER command: ANSWER: B

A) Created
B) Modified
C) Recreated
D) Deleted

37. Which of the following statement is TRUE about SQL Injection? ANSWER: D / A

A) SQL Injection is an Object Penetration Technique
B) SQL Injection is a Function Penetration Technique
C) SQL Injection is a Code Penetration Technique
D) SQL Injection is a Database Penetration Technique

Part 3: Short Answer Questions & Complete Solutions

1. How do you declare a function with parameters in Kotlin? Declare a public function which takes two named integer parameters and returns an integer.

Model Answer:

```
fun addNumbers(num1: Int, num2: Int): Int {
    return num1 + num2
}
```

In Kotlin, functions are public by default. Parameters specify variable name and type (`Int`), followed by return type after colon.

2. Write down the name of different types of Association.

Model Answer:

- **One-to-One (1:1):** One record in Table A connects to exactly one record in Table B.
- **One-to-Many (1:N):** One record in Table A relates to multiple records in Table B.
- **Many-to-One (N:1):** Multiple records in Table A map back to one record in Table B.
- **Many-to-Many (M:N):** Multiple records in Table A relate to multiple records in Table B (resolved via join table).

3. What's the difference between instance method and static method?

Model Answer:

- **Instance Method:** Belongs to an individual object of a class and can access instance variables via `this`. Invoked on an object instance (`obj.method()`).
- **Static Method (Companion Object in Kotlin):** Belongs to the class itself, not tied to any specific instance. Invoked directly on class name (`Class.method()`).

4. What is a constructor in Kotlin, and how many types of constructors are there?

Model Answer: A constructor initializes class properties when an object is instantiated. Kotlin features **2 types** of constructors:

1. **Primary Constructor:** Declared in the class header right after the class name.
2. **Secondary Constructor:** Declared inside the class body using the `constructor` keyword (delegates to `this()`).

5. How do you use a "while" loop to find the sum of even numbers from 1 to 20 in Kotlin?

Model Answer:

```
var sum = 0
var i = 1
while (i <= 20) {
    if (i % 2 == 0) {
        sum += i
    }
    i++
}
// Final sum = 110
```

6. What is an emulator?

Model Answer: An Android Emulator (AVD) is a software simulation tool running on a PC that replicates the hardware architecture, sensors, display, and Android OS of a physical mobile device for testing and debugging applications.

7. How to handle multiple table relationship One-to-One in SQL?

Model Answer: A One-to-One relationship is handled by placing the **Primary Key** of one table as a **Foreign Key** into the related second table, and assigning a `UNIQUE` constraint on that foreign key column.

8. What is the difference between the var and val keywords when declaring variables in Kotlin?

Model Answer:

- `val` (Value): Declares an **immutable** / **read-only** variable (cannot be reassigned once initialized).
- `var` (Variable): Declares a **mutable** variable (value can be updated/reassigned anytime).

9. What is the Activity of an app?

Model Answer: An `Activity` is a fundamental Android component that provides a single focused screen with a user interface through which users interact with the app. It handles lifecycle events like `onCreate()`, `onStart()`, `onResume()`, and `onDestroy()`.

10. What is Action Bar?

Model Answer: The Action Bar (or Toolbar) is a dedicated top app bar UI element that identifies the application title, brand identity, navigation back-button, search actions, and overflow menu items.

11. What is Android Jetpack?

Model Answer: Android Jetpack is a suite of software libraries, tools, and architectural guidance (including ViewModel, LiveData, Room, Navigation, WorkManager) provided by Google to accelerate app development and eliminate boilerplate code.

12. What is the difference between Primary Key and Foreign Key?

Model Answer:

- **Primary Key (PK):** A column that uniquely identifies every individual row in a table (cannot be NULL).
- **Foreign Key (FK):** A column in one table referencing the Primary Key of another table, ensuring referential integrity.

13. SQL Query: Delete row where ID = 1 from "employee" table

Model Answer:

```
DELETE FROM employee WHERE id = 1;
```

14. SQL Query: Show employee names whose salary is less than 30000

Model Answer:

```
SELECT name FROM employee WHERE salary < 30000;
```

Part 4: Oral Assessment / Viva Voce Examination Sheet (with Model Answers)

#	Oral Assessment Question	Expected Model Answer / Key Competency	Result
1	What is ACID in database systems?	**Atomicity** (all or nothing), **Consistency** (valid states), **Isolation** (concurrent transactions don't interfere), and **Durability** (persisted to disk).	[X] Yes
2	Which permission is required to perform network operations in an Android app?	`<uses-permission android:name="android.permission.INTERNET"/>` declared inside the root `AndroidManifest.xml`.	[X] Yes
3	What is the general purpose of HTTP POST and PUT requests?	**POST** is used to submit data to create a new resource on the server. **PUT** is used to replace or update an existing resource idempotently.	[X] Yes
4	How can you handle location updates when an Android app is in the background?	By implementing a **Foreground Service** showing a persistent user notification and holding `ACCESS_BACKGROUND_LOCATION` permission, or using **WorkManager** for deferred periodic location syncs.	[X] Yes
5	What is a Firebase Cloud Messaging (FCM) Token and how is it used?	An FCM Token is a unique registration string assigned to a specific client app instance on a device, used by app servers to direct and deliver targeted push notifications to that exact device.	[X] Yes
6	What is the primary purpose of a Broadcast Receiver in Android?	To listen for, intercept, and respond to system-wide broadcast intents (such as device reboot, battery low, network state change) or custom application broadcast events.	[X] Yes
7	What happens if a registered Broadcast Receiver is not unregistered in an Android App?	It causes serious memory leaks (Activity context leak), potential application crashes during component recreation, and unwanted battery/CPU consumption in the background.	[X] Yes
8	Write down the different types of Services in Android:	1. **Foreground Service** (performs operations noticeable to user with ongoing notification) 2. **Background Service** (performs background tasks not directly noticed by the user) 3. **Bound Service** (offers client-server interface binding components).	[X] Yes
9	How can you handle an incoming phone call audio interruption while music plays in the background?	By requesting and managing **Audio Focus** via `AudioManager` (handling `AUDIOFOCUS_LOSS_TRANSIENT`) and listening to telephony state changes with `TelephonyManager`.	[X] Yes
10	What is the purpose of the "break" statement in a loop in Kotlin?	The `break` statement immediately terminates the execution of the nearest enclosing loop (`for`, `while`, or `do-while`) and jumps execution to the statement immediately following the loop.	[X] Yes

Assessor Assessment & Attitude Checklist

Performance & Attitude Indicators	Satisfactory (Yes)	Not Yet Satisfactory (No)
1. Commitment to Occupational Health and Safety (OHS) & PPE usage	[ ]	[ ]
2. Promptness in carrying out coding and practical setup activities	[ ]	[ ]
3. Sincere and honest in fulfilling assigned assessment tasks	[ ]	[ ]
4. Environmental concern and orderly workstation management	[ ]	[ ]
5. Eagerness to follow specifications and standard coding conventions	[ ]	[ ]
6. Tidiness, punctuality, and respect for peers and assessors	[ ]	[ ]